BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Paper 2: Searching and Sorting

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why should a performance explanation go beyond the word faster?

2. A = 1 and B = 0. What is A AND B?
   A) 1
   B) 2
   C) 0
   D) 10

3. A circuit has X = A AND B, followed by Q = X OR C. For input rows 000, 011, 110, 111, give the pairs (X,Q). Why is a column for X useful?

4. RAM is full. Apps keep moving data between RAM and an SSD. Correct both claims. Explain why. (a) Virtual memory needs no secondary storage. (b) Virtual memory is faster than RAM because SSDs hold more.

Today's objectives

Explain the core ideas of paper 2: searching and sorting using accurate OCR terminology. Apply a clear method to the worked and guided questions.

Key vocabulary - Read, cover, recall

Linear search
Checks elements sequentially, regardless of order.

Binary search
Requires sorted data and repeatedly halves active search range.

Bubble sort
Adjacent comparisons/swaps move large elements right over passes.

Insertion and merge
Insertion grows a sorted prefix; merge sorts sublists before combining.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Linear search
Checks elements sequentially, regardless of order.

Binary search
Requires sorted data and repeatedly halves active search range.

Bubble sort
Adjacent comparisons/swaps move large elements right over passes.

Insertion and merge
Insertion grows a sorted prefix; merge sorts sublists before combining.

I Do - Worked model

Trace bubble-sort first pass of [6,1,4,2].

- Compare 6/1 swap -> [1,6,4,2].
- Compare 6/4 swap -> [1,4,6,2].
- Compare 6/2 swap -> [1,4,2,6].
- Conclude 6 has reached the end.

We Do - Guided practice

1. Use binary search to find 17 in [3,7,11,17,23].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Finish the comparison that detects an out-of-order pair of file sizes.

```
leftGB = 6
rightGB = 1
if ... then
print("Swap")
endif
```

You Do - Independent exam questions

Q2. Give the array after one bubble pass on [4,3,2,1]. [3 marks]

Q3. State the prerequisite for binary search. [1 mark]

Q4. Explain insertion sort in two stages. [3 marks]

Exit Ticket - True or False?

Binary search reliably finds targets in any unsorted list.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. It should identify the relevant characteristic and explain its effect on the stated task.

2. C - 0

3. (0,0), (0,1), (1,1), (1,1). X shows the first gate's output, helping work out Q and find errors.

4. (a) Virtual memory uses secondary storage when RAM is too full. (b) Moving data between RAM and slower secondary storage takes time. More storage space does not mean faster access.

I Do - Explain the model

Compare 6/1 swap -> [1,6,4,2].

Compare 6/4 swap -> [1,4,6,2].

Compare 6/2 swap -> [1,4,2,6].

Conclude 6 has reached the end.

We Do - Model answers

1. Found index3.

Code First - Answer

```
leftGB = 6
rightGB = 1
if leftGB > rightGB then
print("Swap")
endif
```

Expected output: Swap

You Do - Self-mark

Q2: [3,2,1,4].

Q3: The list must be sorted.

Q4: Take next key from unsorted section; shift greater prefix elements right and insert key correctly.

GREEN PEN: Choose the hardest answer. Correct it on lined paper.